



HACK KRMU 5.0

Problem Statement ID : PS8

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AgriChain – Transparent Farm to Consumer Tracking

Project Title: AgriChain Transparent Farm to Consumer Tracking Platform–

a digital platform connecting farmers, collection centres, labs, warehouses and consumers through a single tracking system



Problem

Farmer Visibility

Farmers lose sight of final selling price after submitting produce to collection centres or middle agents.

Consumer Uncertainty

Consumers cannot verify origin, lab testing or how price was calculated.

Root Cause

The primary issue is lack of transparency across the chain, not just middlemen.

Solution

AgriChain creates a digital transparency layer that tracks products from farm to final delivery. Farmers can see how much their product is sold for and understand the cost breakdown. Consumers can track their food using a batch ID and see origin, lab results, pricing details, and delivery status. We are removing opacity from the system.





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Flow of Solution

- Our system follows this flow :

Farmer submits produce at an offline collection center. A unique batch ID is generated. The batch moves to:

Lab Testing → Warehouse → Delivery Partner → Consumer

At each stage, the system updates the status digitally.

Farmers track:

Batch status

Final selling price

Profit margin

Consumers track:

Farm origin

Lab testing details

Price breakdown

Delivery progress

Every step is visible and time-stamped.



Tech Stack & Approach

1

Frontend

Next.js + Tailwind CSS — clean, responsive UI.

2

Backend

FastAPI (Python) — authentication, tracking logic, pricing calculations.

3

Database

SQLite for structured batch and user data storage.



Approach: Centralised system with role-based access — farmer dashboard, consumer tracking portal and admin control panel mirroring the physical supply chain digitally.



Uniqueness & Innovation



Traceability

Digital trace for every batch across stages.



Transparent Pricing

Clear cost breakdown and visible margins for farmers.



Lab Verification

Lab results accessible to consumers for trust and safety.



Farmer Profit Tracking

Farmers can monitor final price and profit margin.

We do not replace the supply chain; we make its economics visible.



Feasibility & Challenges

Feasible

AgriChain is practical because it works as a digital tracking system, not a physical logistics company. We are not building trucks or warehouses.

We are building a system that records and displays information clearly.

The platform can easily be scaled from one district to multiple regions because it is based on structured data and role-based dashboards.

Challenges

In real-world implementation, coordination between farmers, labs, and warehouses would be required.

Data accuracy and trust in updates would also be important.

However, our prototype clearly shows that transparency in the supply chain is technically possible and scalable.



Research & References

- **Foundations**

Study of agricultural supply chain structures and pricing inefficiencies.

- **Theory**

Concepts of information asymmetry in economics applied to farmer–consumer gaps.

- **Safety Models**

Food safety practices and lab verification frameworks inform design.

Prototype demonstrates how transparency can transform value chains.





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Thankyou
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